

Platter Matters!

Here's all you need to enter the world of the Terabyte!

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You've just upgraded your ancient PC—got the fastest processor, truckloads of memory and the best graphics card on the market. But before you can really enjoy the power and speed of your new PC, you will wake up to the realisation that there is hardly any space left on your modest hard drive! Games, music, movies, HD clips, e-books—all this eats into that precious real estate. Most people buy hard drives as an afterthought (“Hmm... I have so much money left over, so I'll buy the biggest drive that fits”)—we don't give them as much love as we do processors or RAM. Fortunately, higher capacity drives are now getting much more affordable.

A couple of years ago, 40 GB was the standard across most PCs, but today we see the momentum shifting towards 160 and even 250 GB. Rightly so—for nearly the same price, these drives offer better features and significantly higher performance. Last year a 750 GB hard disk retailed for Rs 14,000; today you can get a better-performing 750 GB drive for just half the price!

Recent developments such as Perpendicular Recording have vastly improved storage density. Hence, fewer components (platters and heads) are required to manufacture drives. Seagate has introduced single platter drives with capacities ranging from 80 GB to 250 GB. Fewer moving parts mean better speed, lower heat generation and hence longer life. Western Digital has introduced energy-efficient drives labelled “Green Power” or GP. These drives intelligently manage power consumption through smart control of rotation speed (RPM) and head actuator movement. They claim to shave off 4 to 5 watts of power, without any significant performance hits.

In this test, we see the industry also aligning itself toward higher capacity drives. For instance, out of the 17 drives we've tested, the bulk comprises of 500 GB, 750 GB and 1 TB drives. In contrast, last year we had just one each of 750 GB and 1 TB, with 500 GB drives making the bulk. Back then, 320 GB drives were the newest thing in the market, and 500 GB were

for the “elite”. Higher capacity drives now offer better value to the customer—at an asking price of Rs 7,500, a single 750 GB drive makes more sense than three 250 GB or two 500 GB drives.

We put these drives through their paces to see which of them fulfil their promises.

Features Interface

Barring the WD Raptor 150 GB, all other drives are SATA II compliant—that is, their interface speed can go up to 3 Gbps. However, note that the SATA II specification also lists down other features, and it does not exclusively refer to 3 Gbps. Around the completion of our test, WD launched a newer version of their Raptor drive—called the VelociRaptor (300 GB) that supports SATA II. We will review it shortly.

All drives have improved SATA connectors with additional latches, ensuring that the cables don't come out like it used to with the older connectors. Also, all drives offer the option of going back to 1.5 Gbps, if there are compatibility issues with your motherboard's SATA controller. Seagate and Western Digital have jumpers for this purpose. In the case of Hitachi, you need to download the Feature tool to downgrade the interface speed.

Buffers And RPMs

Till last year, most drives would come with 8 MB buffer and a select few would boast of 16 MB.

Seagate's newer drives—7200.11 (7200.9 and 7200.10 are older generation), have an integrated 32 MB buffer, and it seems to benefit on the performance front—even if slightly. Hitachi's 1 TB also has 32 MB, but a 16 MB version is also available. Most Western Digital drives have 16 MB, except for the AV drive, which comes with 8 MB—still reasonable for the daily grind. Seagate's 80 and 160 GB drives were the only drives with 2 MB buffers, and we also see it affecting their performance, especially, when you compare the single-platter drives with their 250 GB counterparts. 7200 RPM is still the speed of choice for most hard drive manufacturers—the Raptor was the only drive whirling at crazy 10,000 RPM. The Green Power drives from Western Digital, dynamically adjust the rotational speed, but we understand the speed varies between 5400 to 7200 RPM.

Performance HD Tach RW

HD Tach RW is the first low-level test, where the hard drives are tested without creating any file system—making the results quite accurate. Although burst speeds are not of practical significance, they give us a glimpse of the drive functioning as a whole. We found that Seagate's 7200.10 750 GB achieves the maximum throughput, followed by its newer counterpart—the Barracuda 7200.11 1 TB. Both these drives reach an astonishing 260 Mbps—quite close to the theoretical limit.

Decision Maker			
Usage	You need	Go For	Recommendation
Entry Level PC	A reliable, reasonably fast, decent capacity hard drive	Any 80 GB SATA drive. Also consider extending your budget by a few hundreds.	Seagate Barracuda 7200.10 250 GB or Western Digital Caviar SE 16 250 GB
Mid-Range PC	A high-capacity, better performing hard drive	A 500 GB SATA drive. If your wallet permits, we recommend a 750 GB drive instead	Seagate Barracuda 7200.11 500 GB or Seagate 7200.11 750 GB or Western Digital RE2 750 GB
Gaming PC	A high-throughput hard drive that can handle continuous and rigorous usage	A high capacity drive with low access time and fast I/O	Seagate Barracuda 7200.11 500 GB or Seagate 7200.11 750 GB or Hitachi Deskstar 1TB
Media centre PC	A quiet but high performing drive	An AV-rated drive that has low acoustics with reasonable performance	Western Digital AV 1TB
Backup Drive	A reliable, high capacity drive	A drive with high interface transfer speeds and proven track record	Hitachi Deskstar 1 TB or Western Digital RE2 750 GB